SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 - ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA1007	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 1 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	28/07/2026		

Code of Conduct

I __________________P.Sonu(192365057)__________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____P.Sonu____________________

1. Introduction

Public libraries play a significant role in providing educational resources, research materials, digital content, and knowledge to people of all age groups. Modern libraries manage thousands of books, magazines, journals, digital resources, and memberships across several branches. As the number of library users and resources increases, manual management becomes difficult, leading to inefficient operations and poor user experience.

Most traditional public libraries still rely on manual record keeping or standalone software applications that are not connected across branches. These systems often result in misplaced books, delayed updates, inaccurate inventory records, and poor communication with library members. Members usually need to visit the library physically to check book availability, reserve books, or renew borrowed books. Library staff spend significant time maintaining records instead of providing quality services.

2. Purpose of the System

The purpose of developing the Smart Public Library Management Platform is to replace the traditional manual library management process with an intelligent cloud-based system capable of managing books, members, reservations, inventory, and reports in real time.

The proposed system aims to automate daily library activities such as book issue, return, reservation, renewal, membership approval, and inventory tracking. It also provides analytical reports that help administrators understand borrowing trends and identify popular books.

3. Business Domain

The Smart Public Library Management Platform belongs to the **Education and Library Information Management** domain.

Libraries today not only manage physical books but also digital books, research journals, multimedia resources, and online memberships. Managing these resources efficiently requires modern software capable of handling large amounts of information across different locations.

The software integrates several business processes including:

- Library Membership Management
- Book Inventory Management
- Digital Resource Management
- Reservation Management
- Circulation Management
- Branch Management
- Analytics and Reporting
- Notification Services

The platform serves educational institutions, government libraries, community libraries, and public library networks.

4. Problem Description

Public libraries experience several operational challenges because of outdated systems and manual processes.

Books are often misplaced because inventory updates are not performed in real time. Library staff manually maintain borrowing records, increasing the chances of errors and inconsistencies. Members frequently visit the library only to discover that a required book is unavailable. There is no efficient online reservation system or automated waiting list.

5. Existing System Analysis

The existing library system primarily depends on manual registers or independent desktop applications installed at each branch. Each library branch maintains its own records without sharing information with other branches.

When members wish to borrow a book, librarians manually verify membership details and update issue registers. Book returns are also recorded manually. If a member wants to reserve a book, the request is usually noted in a register or handled verbally.

Inventory updates occur periodically instead of immediately after every transaction, resulting in inaccurate records. Since there is no centralized database, administrators cannot monitor overall library performance effectively.

Strengths of Existing System

The current system has several advantages despite its limitations.

- Easy to operate.
- Requires minimal technical knowledge.
- Low implementation cost.
- Staff are already familiar with manual procedures.
- Suitable for very small libraries.

Limitations of Existing System

The traditional system suffers from numerous disadvantages.

- Manual data entry increases human errors.
- Books may become misplaced due to inaccurate inventory.
- No real-time synchronization across branches.
- No online reservation system.
- No automated notifications.
- Poor inventory management.
- Limited reporting capabilities.

- No recommendation system.
- High operational workload.
- Slow service during peak hours.

These limitations reduce both operational efficiency and customer satisfaction.

6. Proposed System

The proposed Smart Public Library Management Platform is a distributed cloud-based software application designed to automate all major library operations.

The system stores information about books, members, transactions, reservations, digital resources, and branch inventories in a centralized database that is accessible by all library branches.

7. Objectives of the Proposed System

The major objectives of the Smart Public Library Management Platform are:

- Automate book issue and return processes.
- Track library inventory in real time.
- Provide online reservation facilities.

8. Scope of the System

The scope of the Smart Public Library Management Platform includes all essential library management activities.

The system manages member registration, authentication, book catalog management, digital resources, reservations, borrowing, returning, renewal, inventory tracking, notifications, recommendation systems, analytics, and report generation.

9. Stakeholder Analysis

Stakeholder	Roles	Responsibilities
Library Member	Primary User	Search books, reserve books, borrow books, renew books, receive notifications
Librarian	System Operator	Manage books, issue books, return books, update inventory, approve memberships
Library Administrator	Manager	Generate reports, monitor branches, analyze performance, manage users

Stakeholder	Roles	Responsibilities
System Administrator	Technical Support	Database maintenance, security, backup, server monitoring
Government Authority	Owner	Approve policies, funding, monitor performance
Book Supplier	External Stakeholder	Supply new books based on demand reports

10. System Overview

The Smart Public Library Management Platform follows a centralized cloud architecture that connects all library branches using a shared database.

Whenever a member borrows or returns a book, the inventory is updated instantly across all branches. Members can use mobile applications or web browsers to search books, reserve unavailable titles, renew borrowed books, and access digital resources.

11. Functional Requirements

Functional requirements describe the services and operations that the Smart Public Library Management Platform must perform. The system should provide the following functionalities:

FR1: User Registration

The system shall allow new users to register by entering personal details such as name, email address, phone number, and password. After successful registration, a unique Member ID shall be generated.

FR2: User Login

The system shall authenticate registered users using their username/email and password. Different dashboards shall be displayed based on the user's role (Member, Librarian, or Administrator).

FR3: Membership Management

The librarian shall be able to approve, reject, renew, suspend, or cancel library memberships.

FR4: Book Catalog Management

The librarian shall add, edit, update, delete, and categorize books. Each book record shall include ISBN, title, author, publisher, category, edition, and availability status.

FR5: Book Search

Members shall search books using title, author, ISBN, category, publisher, or keywords. Search results shall display availability and branch location.

FR6: Book Borrowing

The librarian shall issue available books to members after validating their membership status. The system shall automatically record the borrowing date and due date.

FR7: Book Return

The librarian shall record returned books, update the inventory, and calculate overdue fines if applicable.

FR8: Book Renewal

Members shall renew borrowed books online if no reservation exists for the selected book.

FR9: Book Reservation

Members shall reserve books that are currently unavailable. The system shall maintain a waiting list and notify users when the book becomes available.

FR10: Digital Resource Access

Members shall access digital resources such as e-books, research journals, magazines, and audiobooks through the platform.

FR11: Notification Management

The system shall automatically send email or SMS notifications for due dates, overdue books, reservation availability, membership expiry, and newly added books.

FR12: Recommendation Engine

The platform shall recommend books by analyzing borrowing history, favorite categories, and reading patterns.

FR13: Demand Prediction

The system shall predict future demand for books by analyzing historical borrowing data and seasonal trends.

FR14: Inventory Management

The system shall track book availability across all library branches in real time and support inventory transfers between branches.

FR15: Report Generation

The administrator shall generate reports on book circulation, overdue books, inventory utilization, member activities, branch performance, and popular titles.

FR16: Branch Management

The administrator shall manage multiple library branches, including adding, updating, or removing branch details.

FR17: Fine Management

The system shall calculate overdue fines automatically and maintain payment records.

FR18: Role-Based Access Control

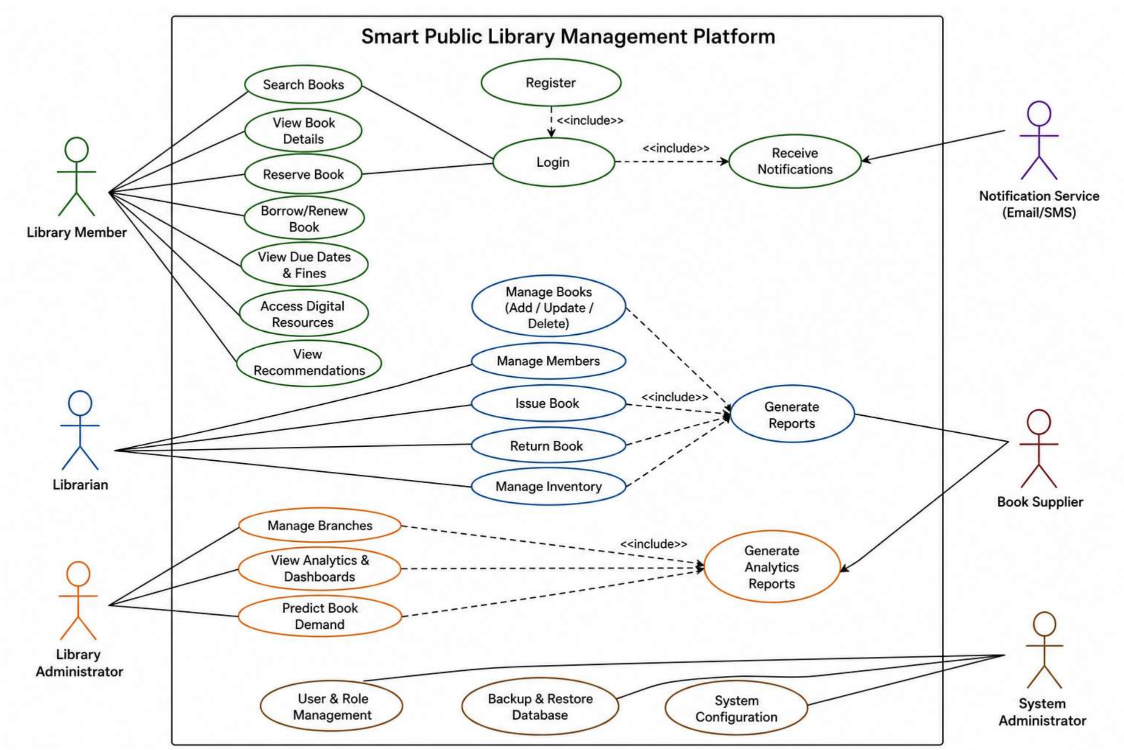
The system shall provide different permissions for members, librarians, administrators, and system administrators.

FR19: Backup Management

The system administrator shall perform scheduled database backups and recovery operations.

FR20: Audit Log

The system shall maintain logs of all important activities for monitoring and security purposes.



Completeness Check

All major stakeholder requirements have been included.

Requirement	Status
User Registration	Completed
Login	Completed
Membership Management	Completed
Book Search	Completed

Requirement	Status
Issue & Return	Completed
Reservation	Completed
Digital Resources	Completed
Notifications	Completed
Recommendation	Completed
Reports	Completed
Analytics	Completed
Inventory Tracking	Completed

Use Case 1 – Register Member

Actor: Library Member

Description:
A new user creates a library account to access library services.

Pre-condition:
The user is not already registered.

Main Flow:

1. Open the registration page.
2. Enter personal information.
3. Create username and password.
4. Submit the registration form.
5. System validates the information.
6. Member account is created successfully.

Post-condition:
The new member can log in to the system.

Use Case 2 – Search Book

Actor: Library Member

Description:
The member searches for books available in the library.

Pre-condition:
The member has logged in.

Main Flow:

1. Enter search keyword.
2. System searches the database.
3. Matching books are displayed.
4. Member views availability and branch location.

Post-condition:
Member selects a book for borrowing or reservation.

Use Case 3 – Issue Book

Actor: Librarian

Description:
The librarian issues an available book to a registered member.

Pre-condition:
The member has an active membership.

Main Flow:

1. Search member details.
2. Scan book barcode.
3. Verify book availability.
4. Issue the book.
5. Update inventory.
6. Generate due date.

Post-condition:
Book status changes to **Issued**.

Use Case 4 – Reserve Book

Actor: Library Member

Description:
Members reserve books that are currently unavailable.

Pre-condition:
Book is already borrowed by another member.

Main Flow:

1. Search unavailable book.
2. Click **Reserve**.
3. Reservation request stored.
4. Waiting list updated.

Post-condition:
Member receives notification when the book becomes available.

Use Case 5 – Generate Report

Actor: Library Administrator

Description:
Administrator generates reports about library activities.

Pre-condition:
Administrator is logged in.

Main Flow:

1. Select report type.
2. System collects data.
3. Generate charts and statistics.
4. Export report as PDF or Excel.

Post-condition:
Report is stored and available for future reference.

Conclusion:

The Smart Public Library Management Platform is an advanced cloud-based library management solution designed to modernize public library operations. It addresses the limitations of traditional manual systems by integrating real-time inventory tracking, online reservations, digital resource management, intelligent recommendations, demand prediction, automated notifications, and comprehensive reporting into a single platform.

The Software Requirement Specification (SRS) presented in this report defines the project's objectives, scope, stakeholders, functional requirements, non-functional requirements, use cases, data requirements, external interfaces, assumptions, constraints, and validation process. These requirements provide a clear roadmap for developers, testers, project managers, and stakeholders throughout the software development life cycle.

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